

Section C

10 **(short)** A capacitor is formed by two parallel plates that are spaced by a distance d , and each carries a charge of $+Q$ and $-Q$, respectively. Both plates have an area of A . Now the distance between the two plates is enlarged from d to $2d$, as shown in Fig. 5.

- (a) What is the change of the stored electrostatic energy? [5]
- (b) What is the initial force used to enlarge the separation of the two plates at distance d ? [5]

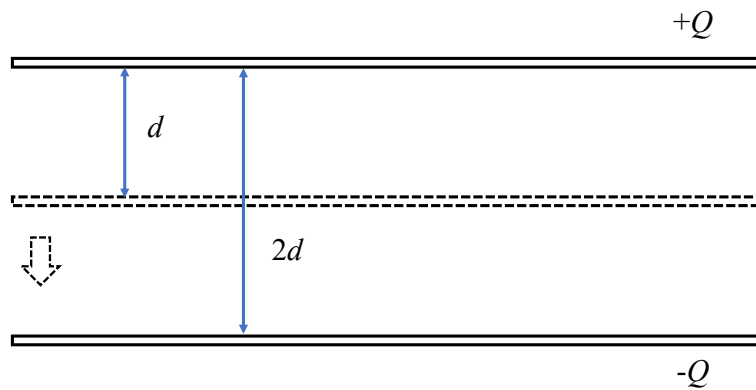


Fig. 5